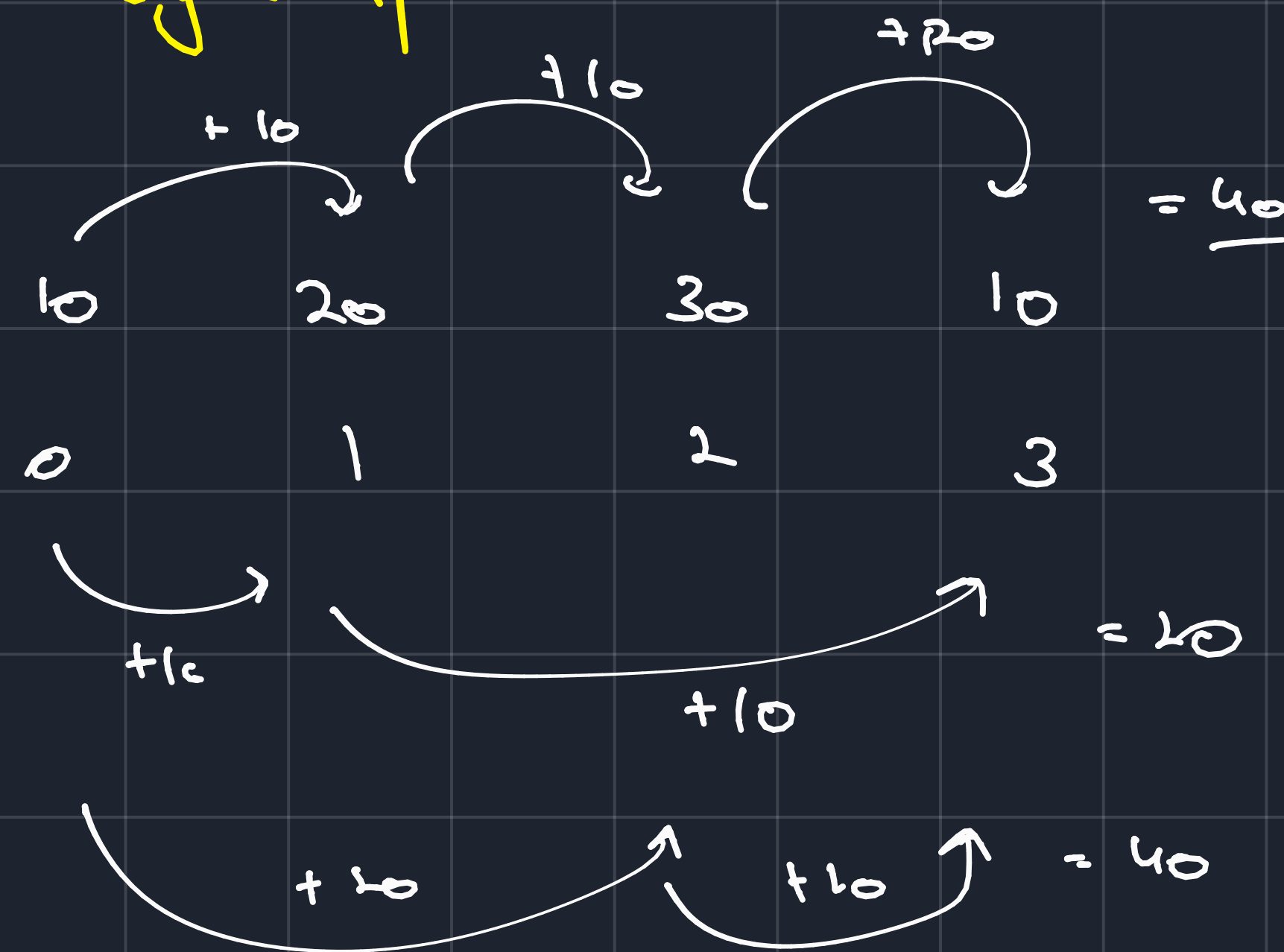
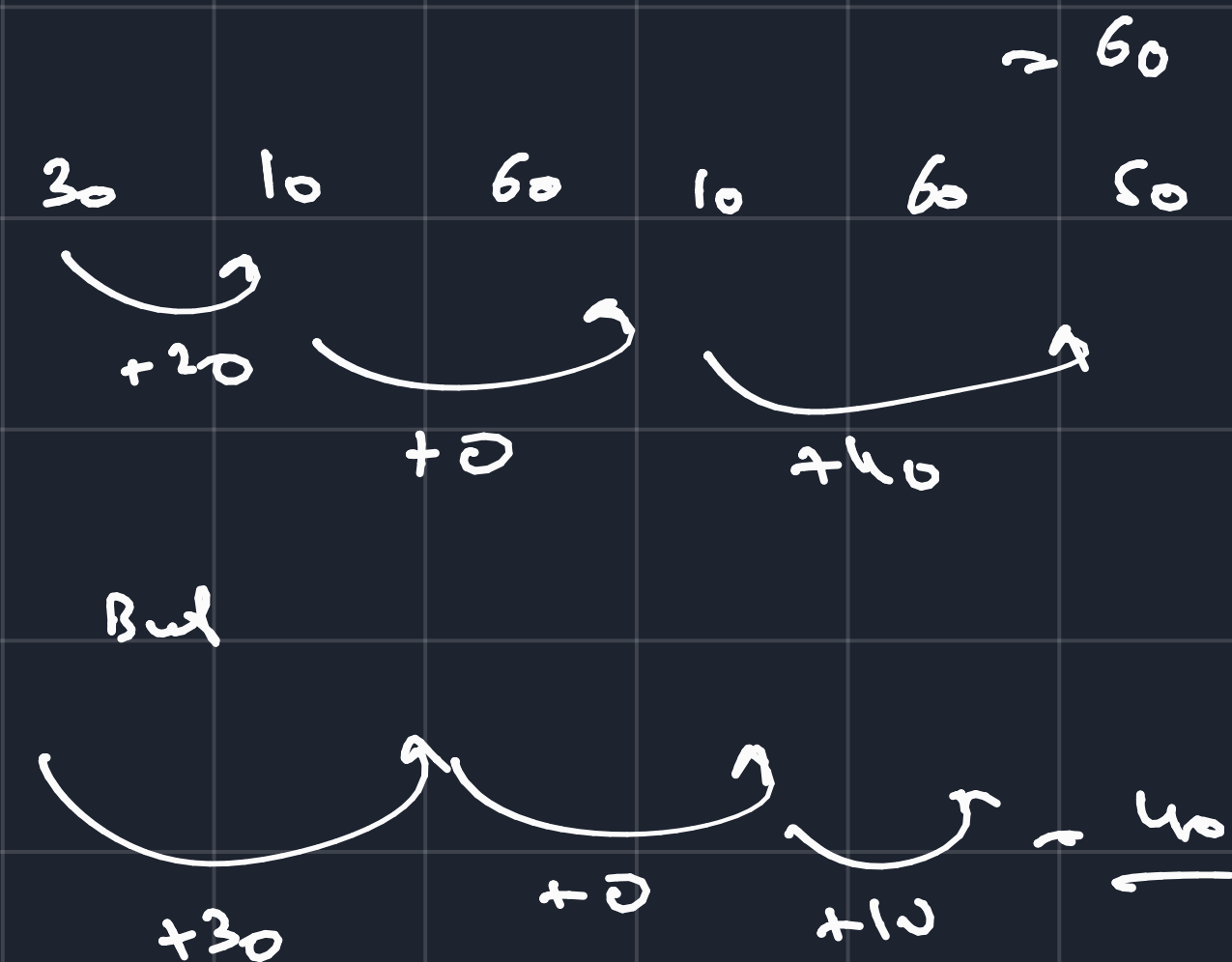


↳ Frog Jump.



↳ why a greedy doesn't work.



↳ $f(n-1) \rightarrow$ min energy req to reach $n-1$ from

0

↳ $f(ind)$

f

if (ind == 0) return 0;

left = $f(ind-1) + \text{abs}(a[ind] - a[ind-1])$

if (ind > 1)
right = $f(ind-2) + \text{abs}(a[ind] - a[ind-2])$

return min(left, right)

}

how many
calls?

↳ 1, 2, 3

1, 2

1, 2, 3

[30, 10, 60, 10, 60, 50], n=6

$f(5) = \underline{40}$

$f(4) = 30$

$f(3)$

20

f

$f(3)$

$f(2)$

abs

$a[2]$

$-a[1]$

overlapping

subproblems

$f(2) = 20$

$f(1) = 20$

$f(1) = 20$

$f(0)$

$f(0)$

↳ $f(1) + 50 = 80$

↳ $20 + 30 = 50$

$P(1)$

20 +

f

$f(1) + \text{abs}(2-1)$

$20 + 1 = 21$
 $20 + 30 = 50$

↳ memoization → top-down

↳ look at the number of changing

parameter - .s create $dp[\text{max}+1]$

↳ Before returning, store it.

↳ check if it's previously computed

↳ 3rd

↳ for ($i=1 \rightarrow n-1$)

f.s = $dp[\text{ind}-1] + \text{abs}(a[\text{ind}] - a[\text{ind}-1])$

if ($\text{ind} > 1$)

ss = $dp[\text{ind}-2] + \text{abs}(\text{ind} - \text{ind}-2)$

$dp[\text{ind}] = \min(f.s, s.s)$ }

T.C → $O(N)$

S.C → $O(w) + O(w)$

↳ Tabulation

↳ Bottom-up.

Steps

↳ int $dp[n] \rightarrow -1$

↳ Look for the base case.

↳ write what in dp

↳ $dp[0] = 0$